SPRING CLOUD – 40 Deep Interview Questions (3-Line Answers)

1. What is Spring Cloud?

Spring Cloud provides tools to build distributed systems and microservices.

It handles service discovery, configuration, and fault tolerance.

It simplifies cloud-native application development.

2. What is Eureka Server?

Eureka Server is a service registry.

Microservices register themselves on startup.

Other services discover them dynamically.

3. What is Eureka Client?

Eureka Client registers the service with Eureka Server.

It fetches registry information periodically.

Used for dynamic service discovery.

4. How does service registration work in Eureka?

When service starts, it sends heartbeat to Eureka Server.

Registry stores service metadata and instance details.

If heartbeat stops, instance is removed.

5. What is Self-Preservation mode in Eureka?

Prevents mass removal of services during network failure.

Stops expiring instances if heartbeat drops drastically.

Improves resilience during outages.

6. What is Spring Cloud Config Server?

Config Server provides centralized configuration management.

Stores properties in Git or external repository.

Microservices fetch configuration at startup.

7. What is Config Client?

Config Client retrieves configuration from Config Server.

Supports dynamic refresh.

Avoids duplication of configuration files.

8. What is @RefreshScope?

Allows refreshing beans without restarting application.

Used with Actuator refresh endpoint.

Useful for dynamic config updates.

9. What is Spring Cloud Gateway?

Gateway is API Gateway built on Spring WebFlux.

Handles routing and filtering.

Supports rate limiting and authentication.

10. What are Gateway Filters?

Filters modify requests and responses.

Can be pre-filter or post-filter.

Used for logging, authentication, and transformation.

11. What is Route Predicate?

Defines routing conditions in Gateway.

Based on path, header, or method.

Determines which service receives request.

12. What is Load Balancer in Spring Cloud?

Distributes traffic across service instances.

Spring Cloud LoadBalancer replaces Ribbon.

Improves availability and scalability.

13. What is Feign Client?

Feign is declarative REST client.

Simplifies inter-service communication.

Integrated with service discovery.

14. How does Feign integrate with Eureka?

Feign automatically resolves service name via Eureka.

No hardcoded URLs required.

Supports client-side load balancing.

15. What is Resilience4j?

Resilience4j is fault tolerance library.

Provides circuit breaker and retry mechanisms.

Replaced Hystrix in modern systems.

16. What is Circuit Breaker in Spring Cloud?

Prevents repeated calls to failing service.

Switches to fallback method when failure occurs.

Improves system stability.

17. What is Retry mechanism?

Retries failed service calls automatically.

Configured with max attempts and delay.

Used with circuit breaker.

18. What is TimeLimiter?

Limits maximum execution time of service call.

Prevents long blocking operations.

Enhances resilience.

19. What is Rate Limiting in Gateway?

Limits number of requests per user.

Prevents abuse or DDoS attacks.

Configured via Redis or in-memory store.

20. What is Distributed Tracing?

Tracks request flow across microservices.

Helps identify bottlenecks.

Implemented using Sleuth and Zipkin.

21. What is Spring Cloud Sleuth?

Adds trace and span IDs to logs.

Helps correlate logs across services.

Works automatically with Spring Boot.

22. What is Zipkin?

Zipkin collects and visualizes trace data.

Displays request timeline across services.

Useful for performance monitoring.

23. What is Config Refresh endpoint?

Actuator endpoint to reload config at runtime.

Used with @RefreshScope.

Avoids service restart.

24. What is Bootstrap Context?

Loads configuration before main ApplicationContext.

Used in older Spring Cloud versions.

Replaced by Config Data API in newer versions.

25. What is Service-to-Service Security?

Uses OAuth2 or JWT tokens.

Token validated in downstream services.

Ensures secure communication.

26. What is API Gateway vs Service Mesh?

Gateway handles external traffic.

Service mesh manages internal traffic.

Mesh provides advanced traffic control.

27. What is Config Encryption?

Encrypts sensitive properties in config files.

Decrypted at runtime.

Prevents credential exposure.

28. What is Hystrix?

Hystrix was Netflix fault tolerance library.

Provided circuit breaker pattern.

Now replaced by Resilience4j.

29. What is Client-Side Load Balancing?

Client selects service instance.

Works with Eureka registry.

Improves performance.

30. What is Server-Side Load Balancing?

Load balancer distributes requests externally.

Examples include Nginx or AWS ELB.

Independent of application code.

31. What is Gateway Authentication Filter?

Validates JWT before forwarding request.

Prevents unauthorized access.

Centralizes security logic.

32. What is Spring Cloud Bus?

Broadcasts config changes across services.

Uses message broker like Kafka.

Synchronizes configuration updates.

33. What is DiscoveryClient?

Interface to fetch registered services.

Used programmatically for discovery.

Abstracts service registry implementation.

34. What is Heartbeat mechanism?

Service sends periodic heartbeat to registry.

Ensures instance is alive.

Removed if heartbeat fails.

35. What is Failover?

Switching to backup service instance.

Happens automatically via load balancer.

Improves availability.

36. What is Rolling Update with Spring Cloud?

Deploy new instances gradually.

Old instances removed after verification.

Ensures zero downtime.

37. What is Service Metadata?

Extra information stored in registry.

Includes version and environment details.

Used for routing decisions.

38. What is Multi-Region Deployment?

Deploy services in multiple regions.

Improves availability and latency.

Requires global load balancing.

39. What is Cloud-Native Application?

Designed for scalability and resilience.

Uses containers and microservices.

Leverages cloud infrastructure.

40. What is Production Readiness in Spring Cloud?

Includes monitoring, tracing, and resilience.

Supports centralized config and discovery.

Designed for distributed enterprise systems.